

Gamma + beta rhythms

Instructor: Mark Kramer

Today

Explore synchronization between two neural populations

Combine our models of gamma & beta

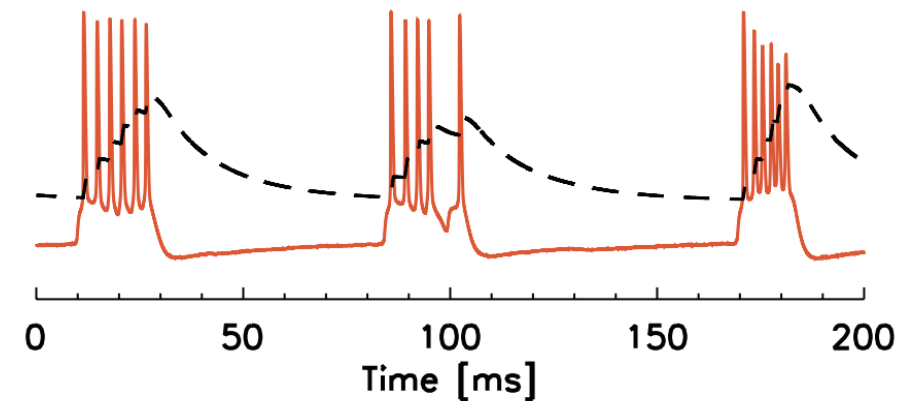
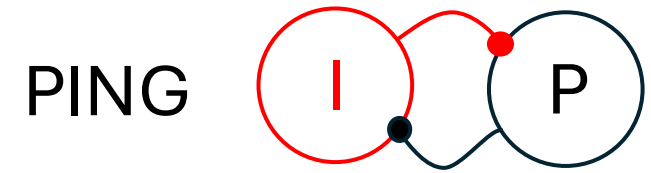
Gamma vs beta rhythms

Gamma (30-80 Hz)

- synaptic interactions important
- link local neural populations

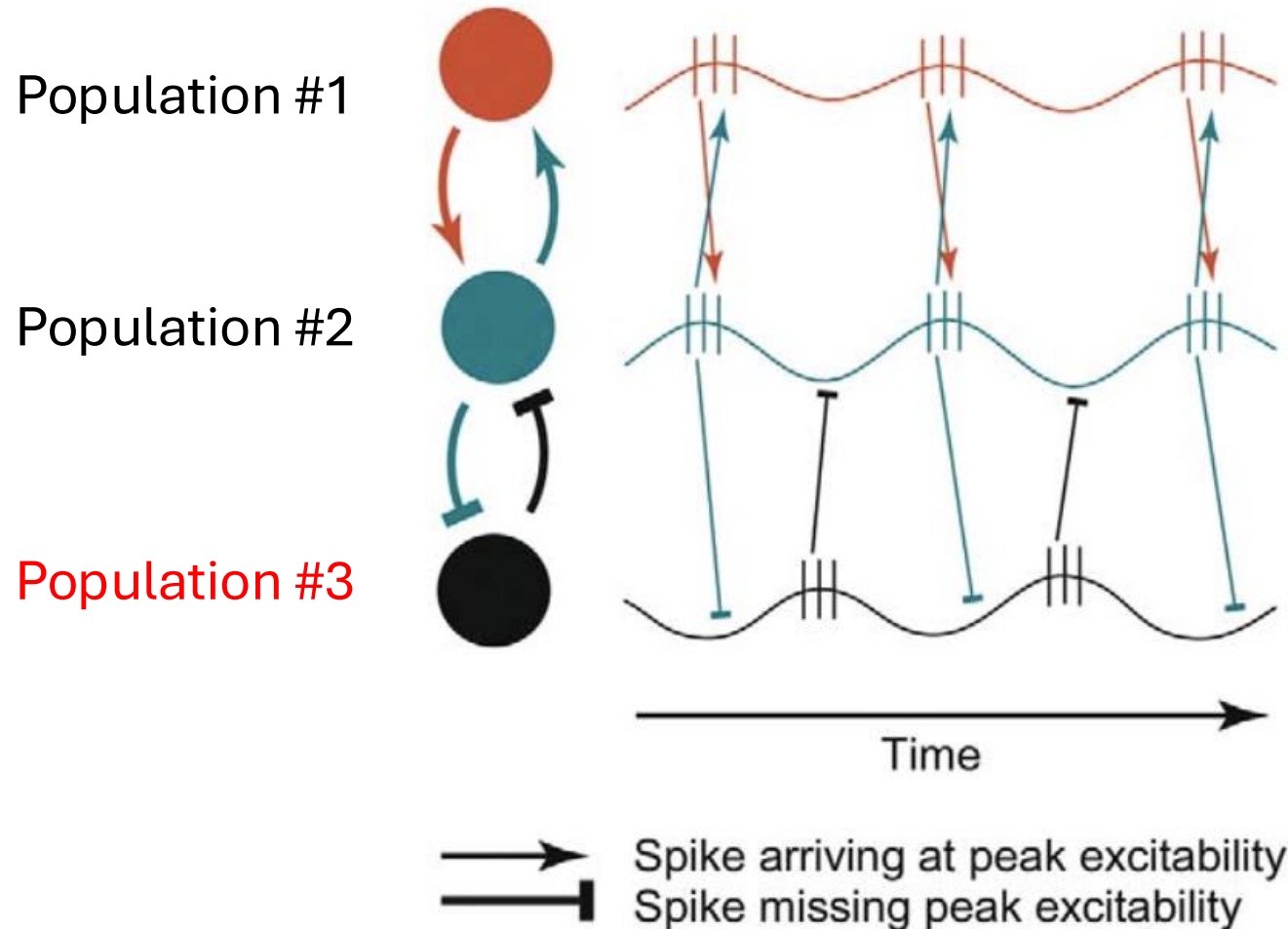
Beta (15-30 Hz)

- membrane currents important
- link distant neural populations



Link neural populations

Q: What does that mean?



Oscillate at the same frequency

Spikes arrive at excitability peaks

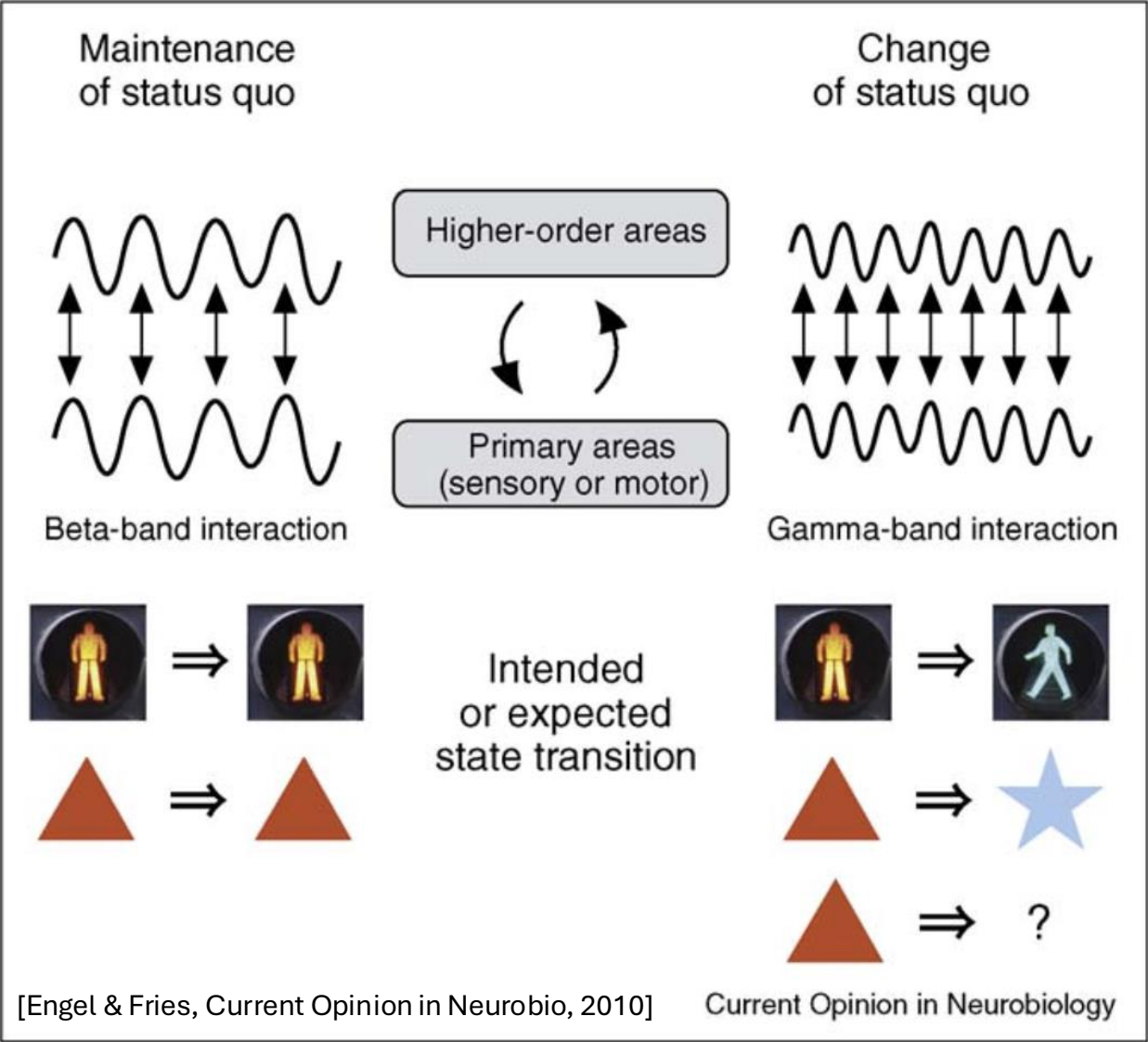
→ Linked

Oscillate at the same frequency

Spikes arrive at excitability troughs

→ Not linked

Gamma vs beta rhythms



Population #1

Linked at beta

Population #2

Population #1

Linked at gamma

Population #2

Q: Simulate it?

Simulation motivation

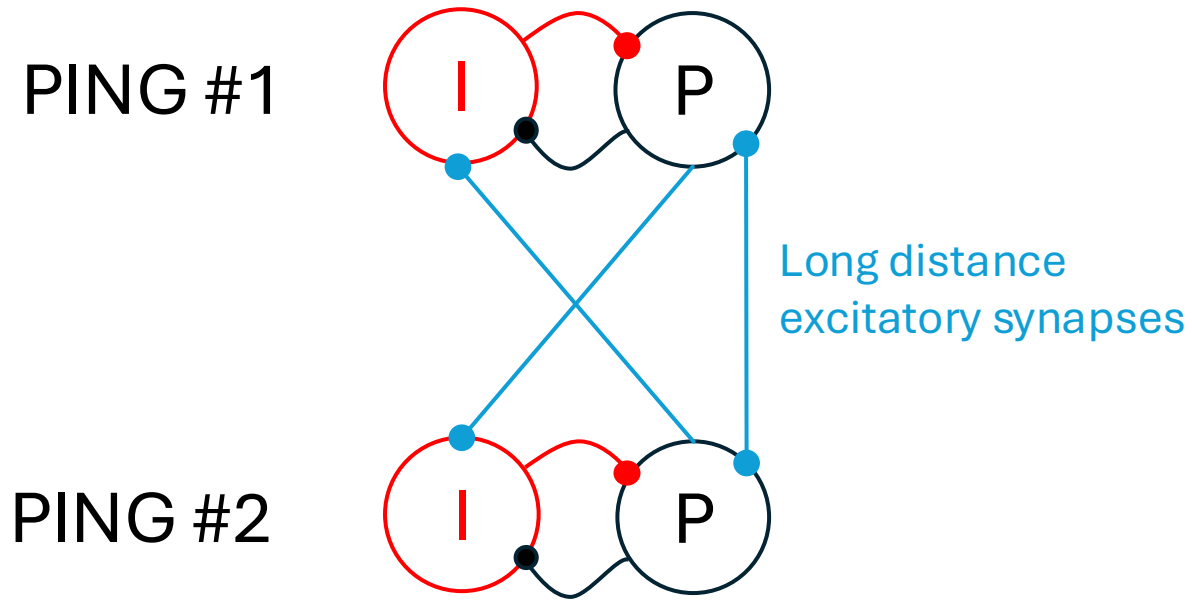
Gamma rhythms and beta rhythms have different synchronization properties

N. Kopell^{†‡}, G. B. Ermentrout[§], M. A. Whittington[¶], and R. D. Traub^{||}

[†]Department of Mathematics and Center for BioDynamics, Boston University, Boston MA 02215; [§]Department of Mathematics, University of Pittsburgh, Pittsburgh PA 15260; [¶]School of Biomedical Sciences, Worsley Building, University of Leeds, Leeds LS2 9NL, United Kingdom; and ^{||}Division of Neuroscience, The Medical School, University of Birmingham, Birmingham B15 2TT, United Kingdom

Remember, theta neuron creators

Consider two PING populations



Encodes color: green

Link (bind) them 

Encodes shape: 

Synchronize activity of the two populations.

Q: How? Connect them

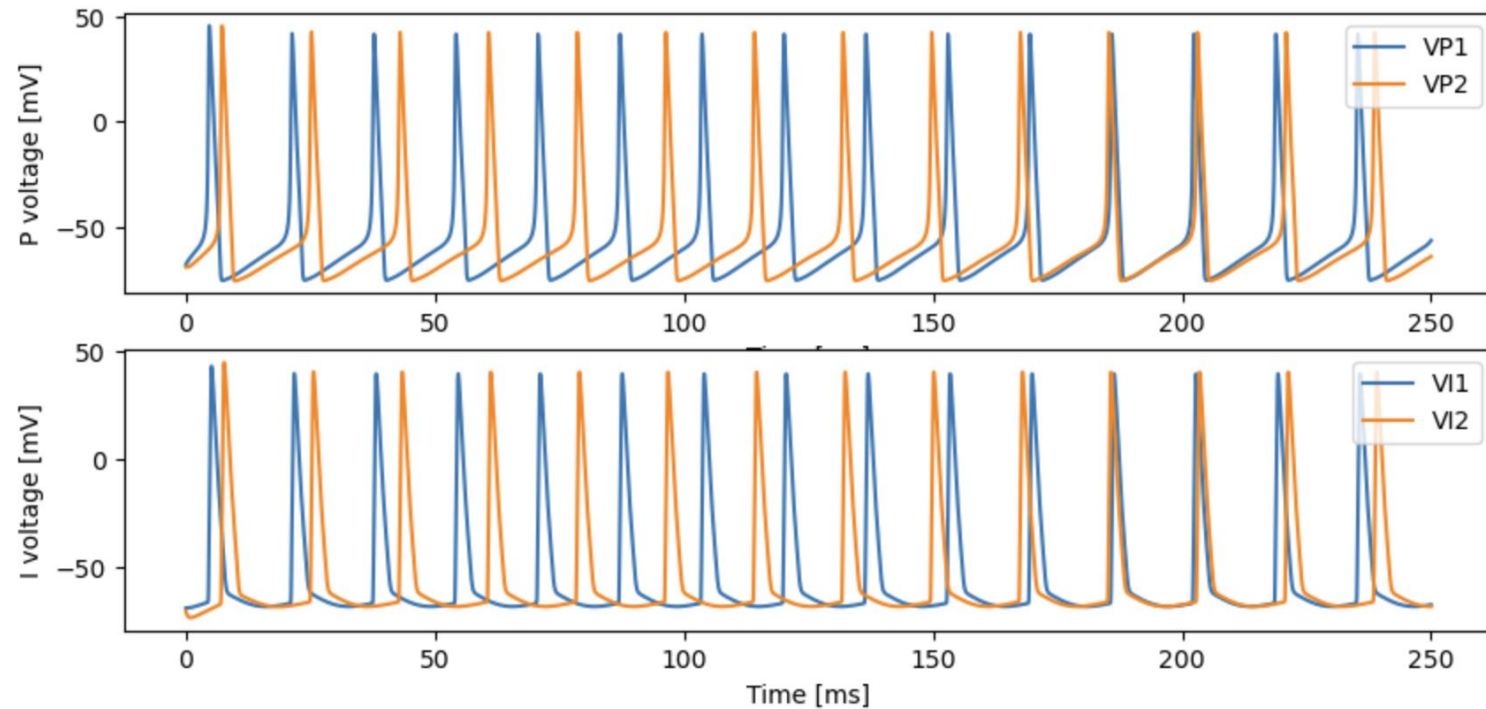
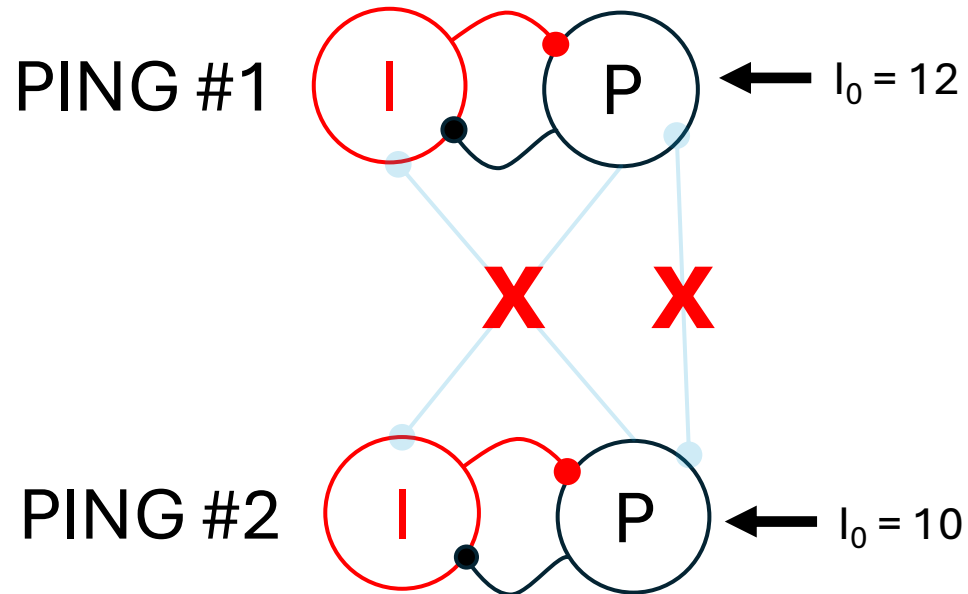
Two PING populations



Python

Two PING populations: Results

Case 1: No long-distance connections, more input to PING #1

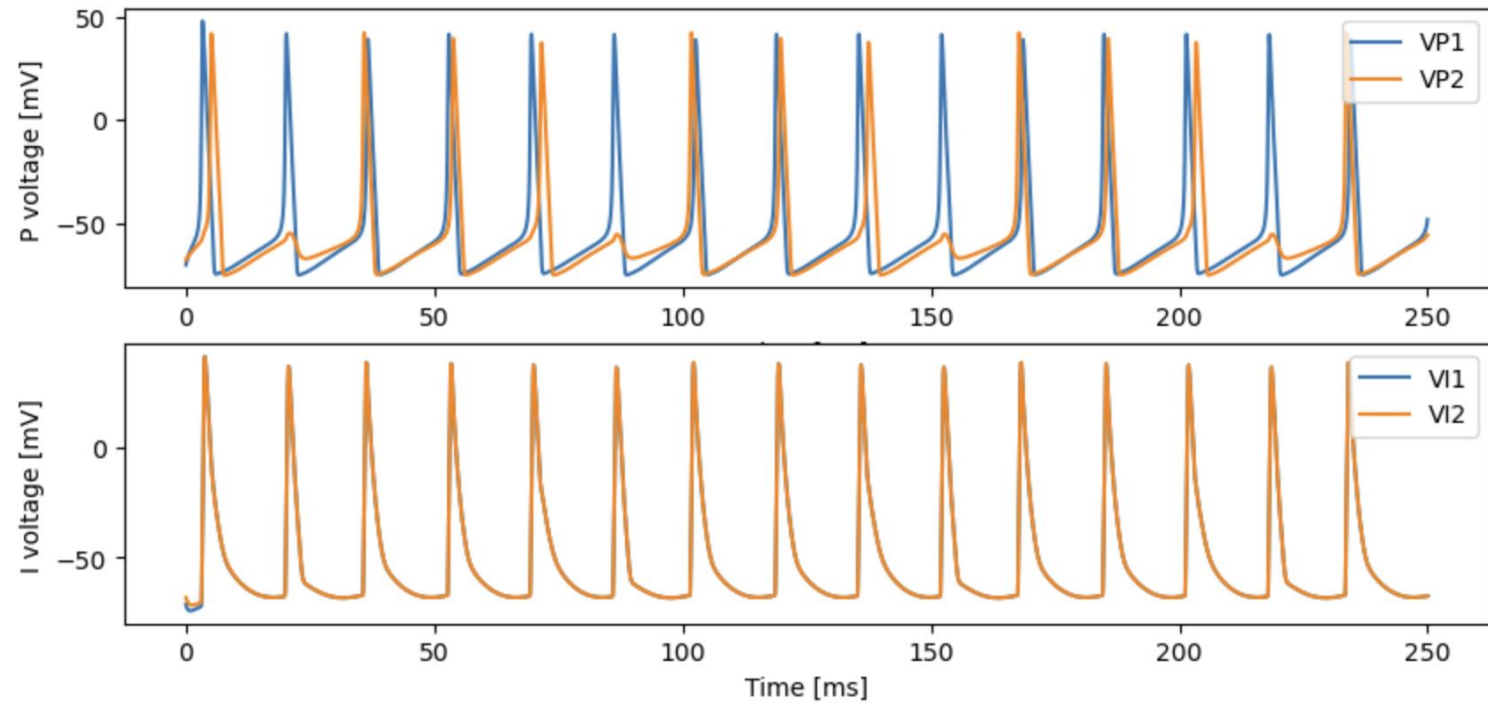
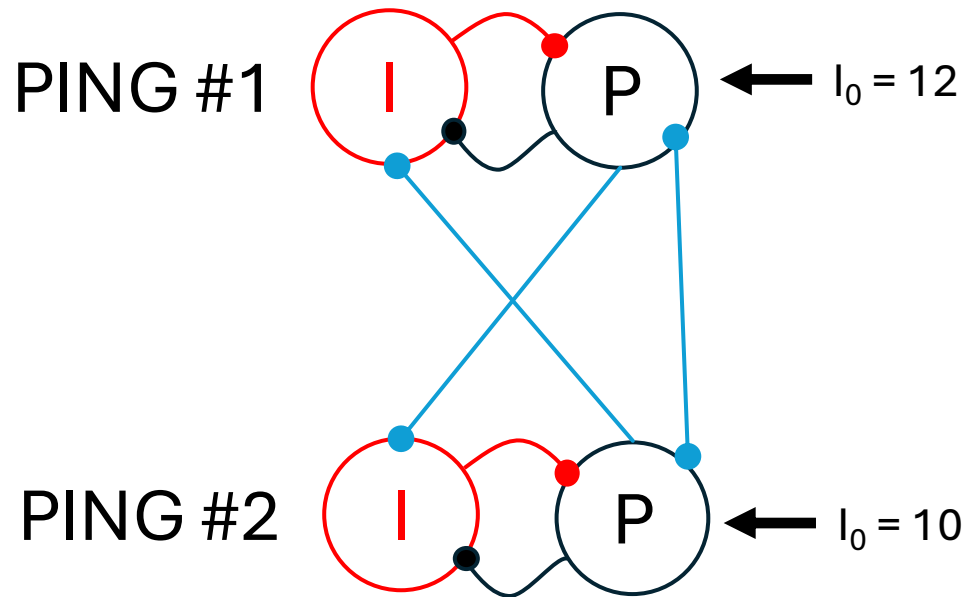


Q: What's happening here?

Q: Are the populations synchronized?

Two PING populations: Results

Case 2: ~~No~~ long-distance connections, more input to PING #1

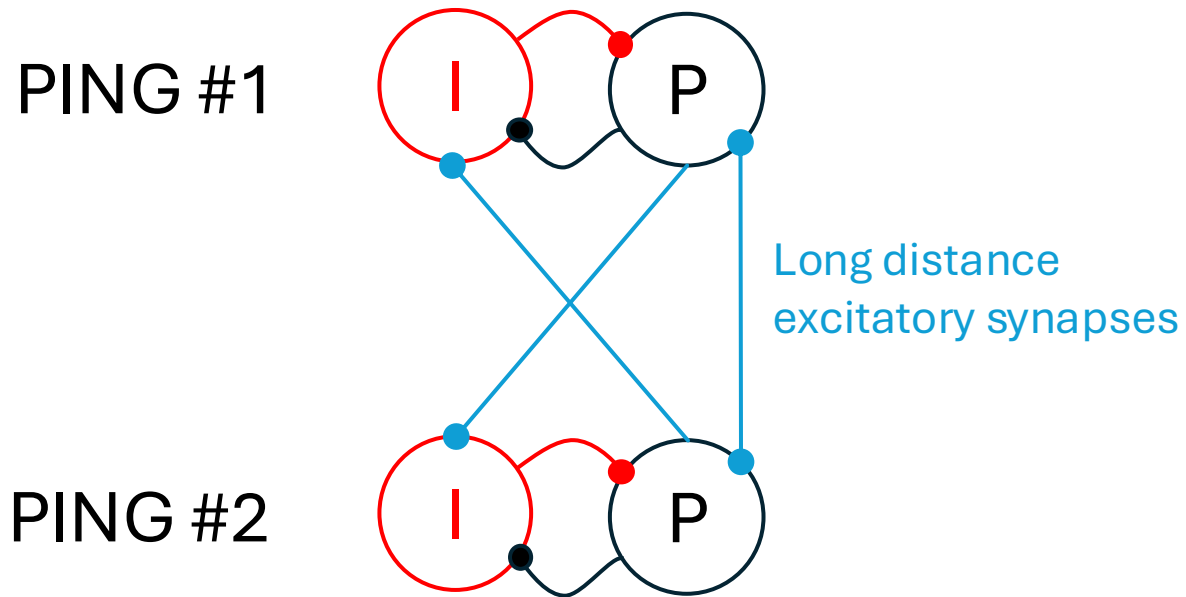


Q: What's happening here?

Q: Are the populations synchronized?

Two PING populations with delays

Synchronized, but missing an important property: **conduction delays**



It takes more time for signals to transit between these distant populations

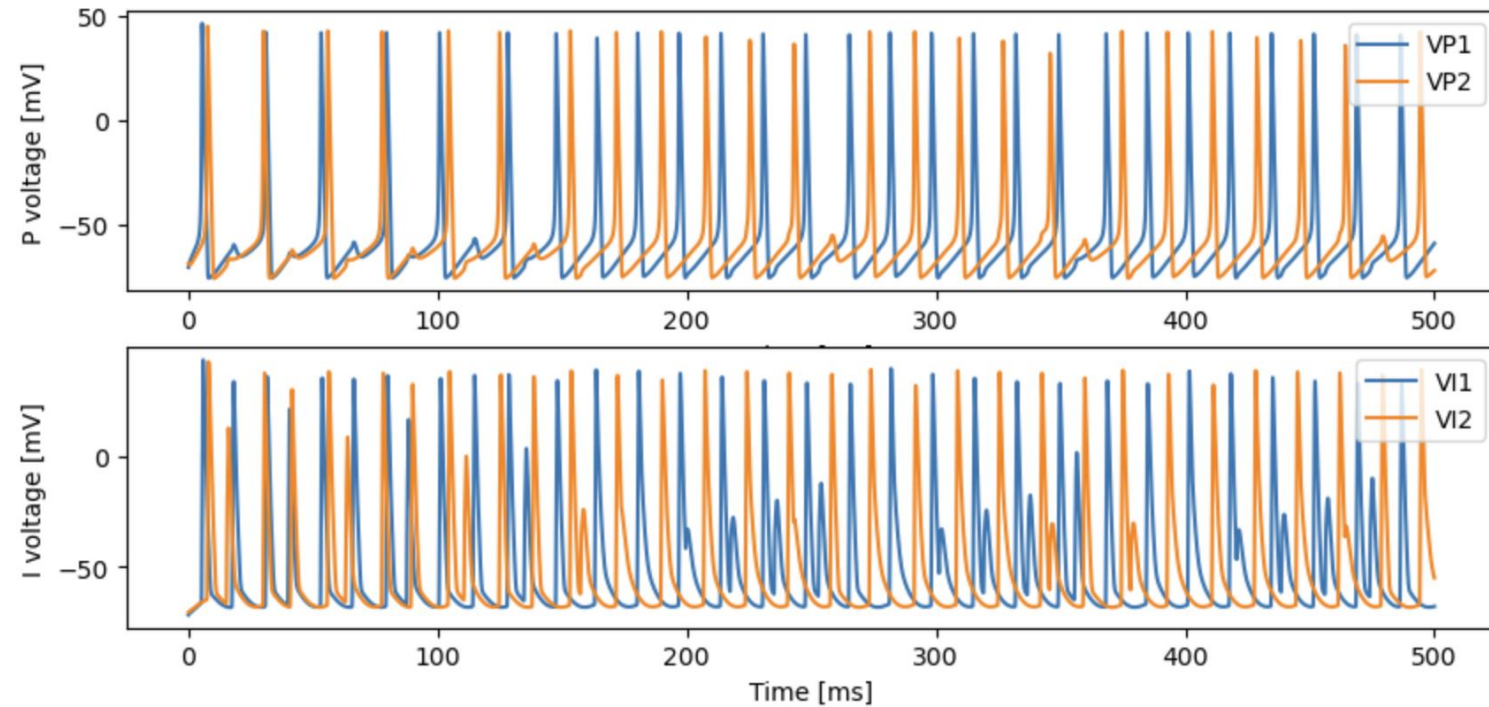
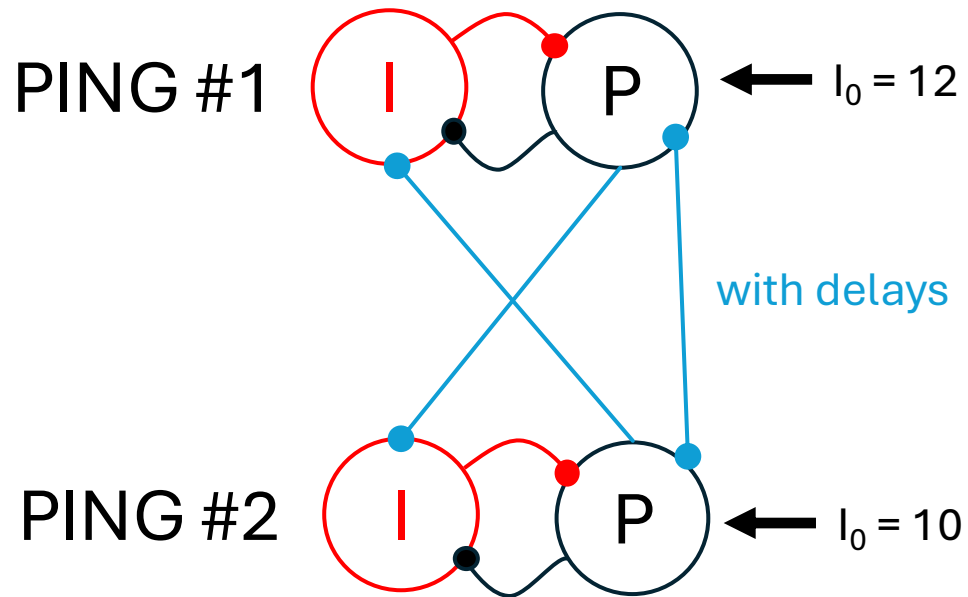
Q: How does this delay impact synchronization?

Two PING populations with delays

Python

Two PING populations with delays: Results

Case 3: Long-distance connections + **delays**, more input to PING #1

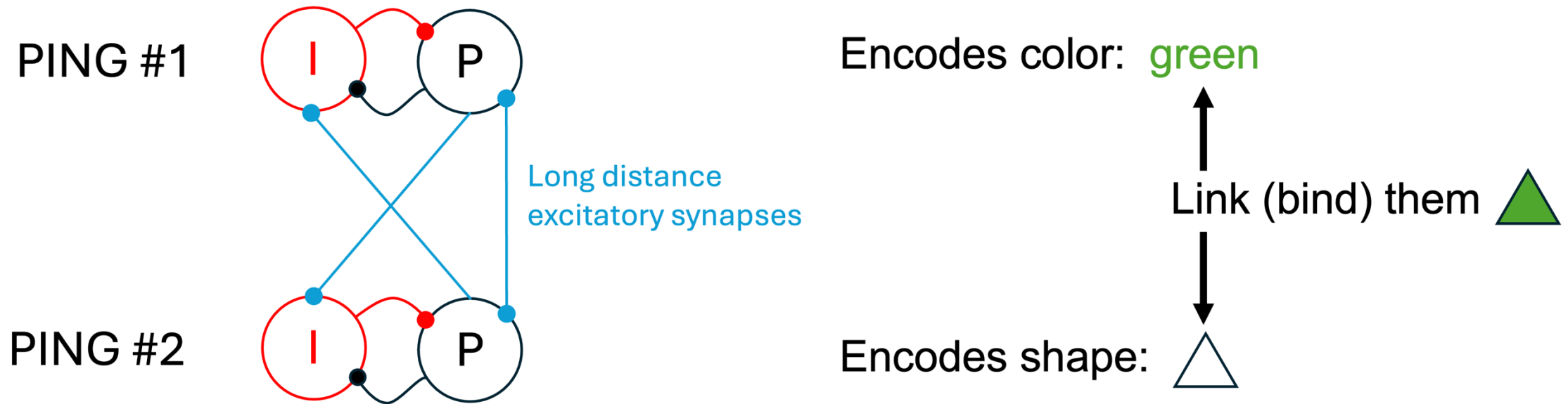


Q: What's happening here?

Q: Are the populations synchronized?

Two PING populations with delays: Results

Wait ... I thought gamma was a synchronizing rhythm?



But, synchronization fails when we include more biological realism.

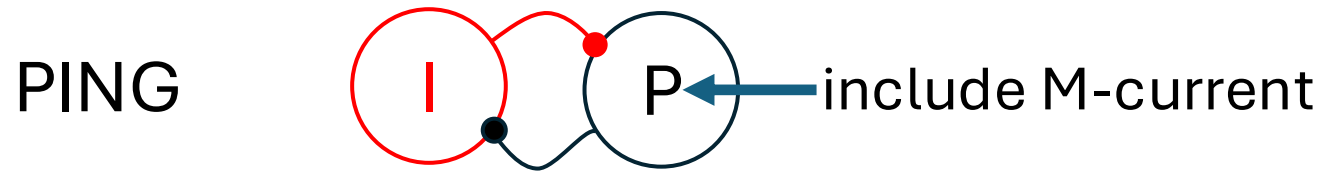
Q: Now what?

Gamma + beta model

Update the model to support both gamma & beta

Gamma: PING

Beta: M-current



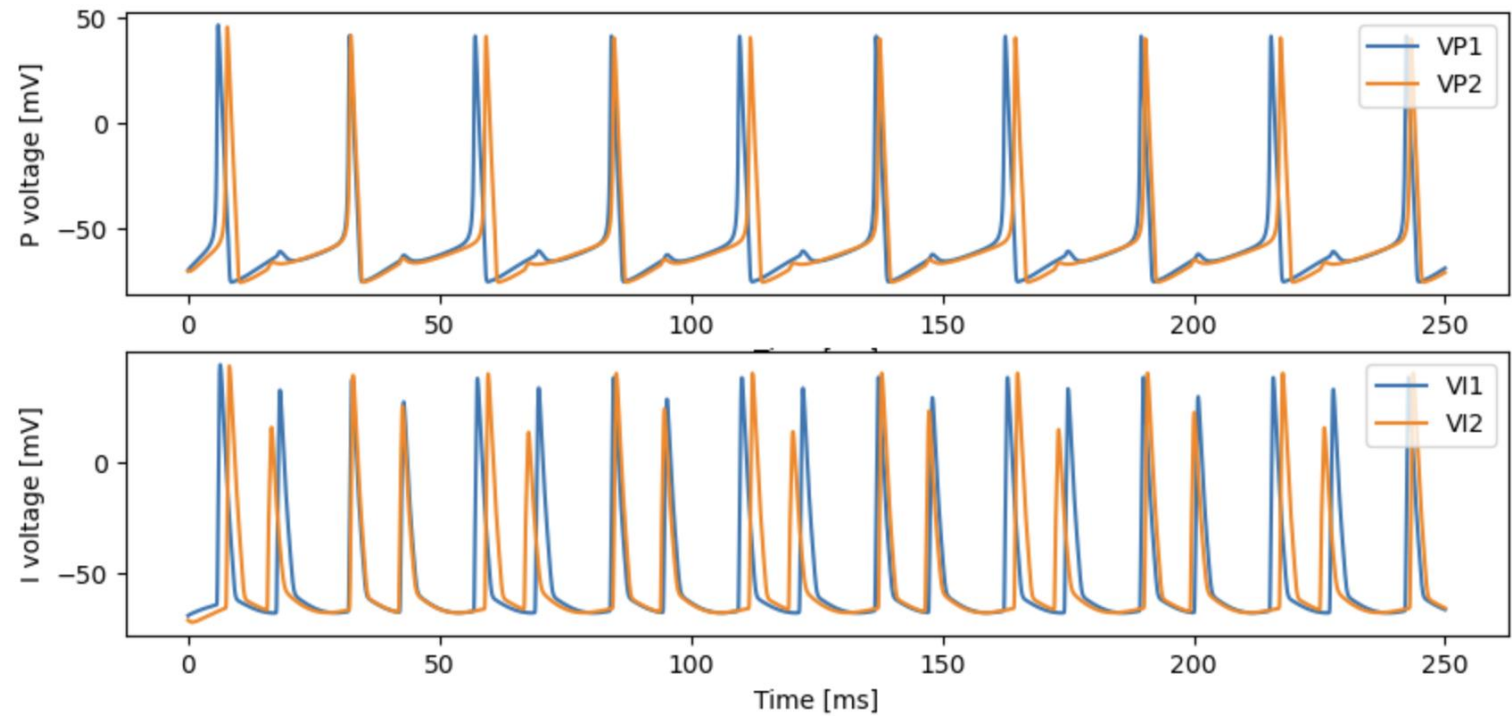
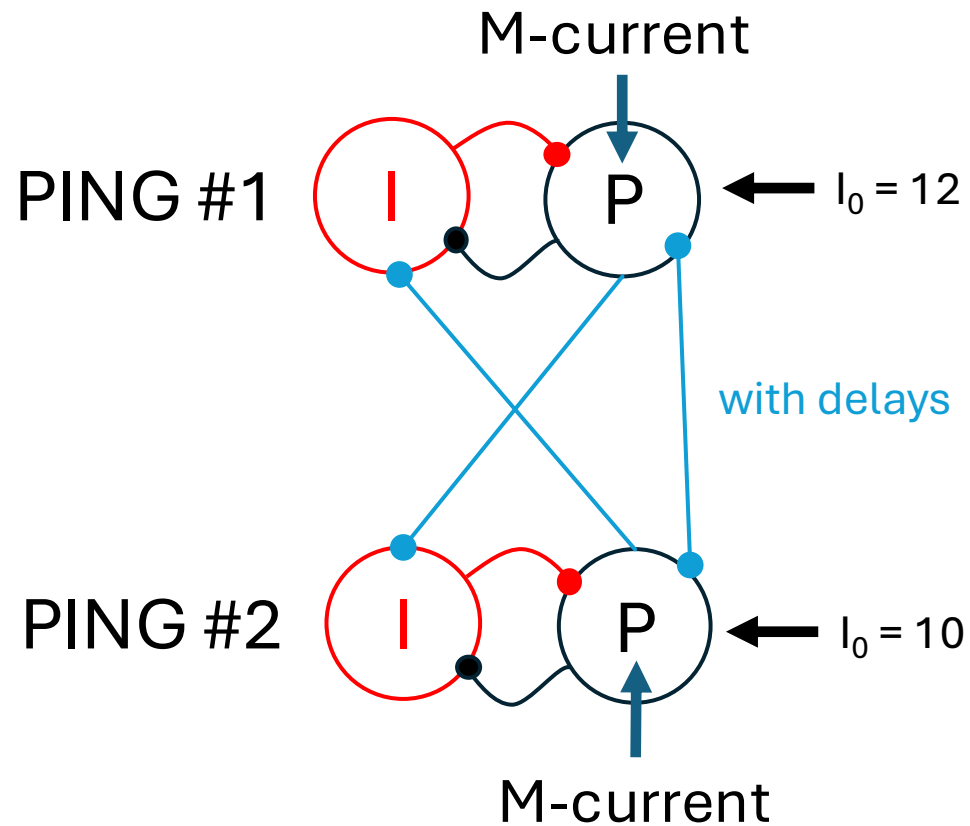
Q: How does including beta impact synchronization?

Gamma + beta model

Python

Gamma + beta model: Results

Case 4: Long-distance connections + delays + M-current, more input to PING #1

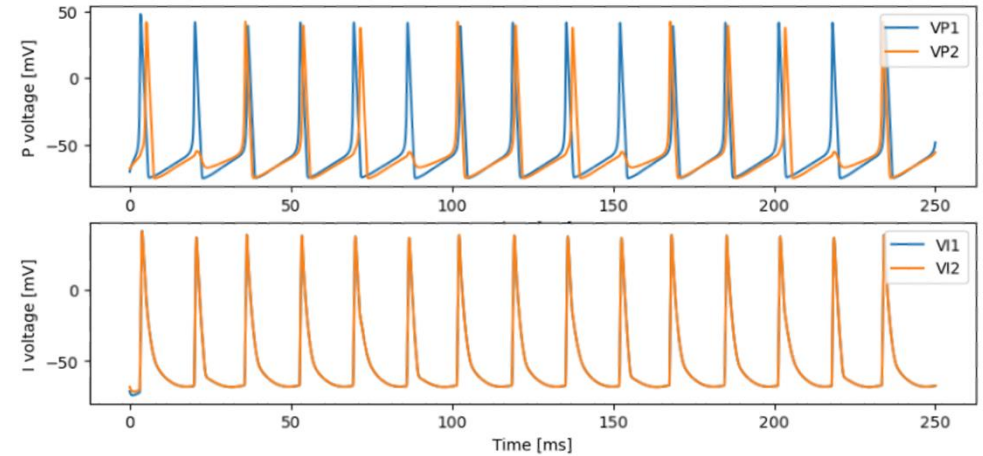
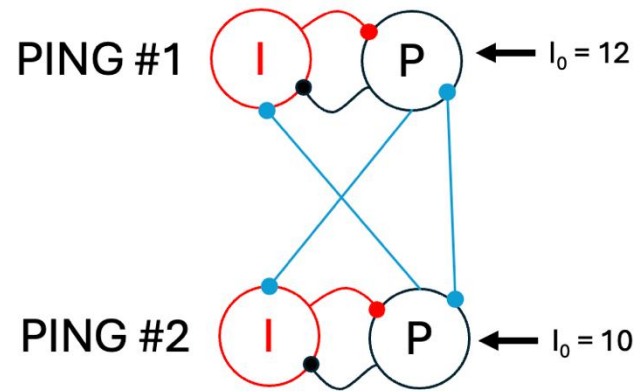


Q: What's happening here?

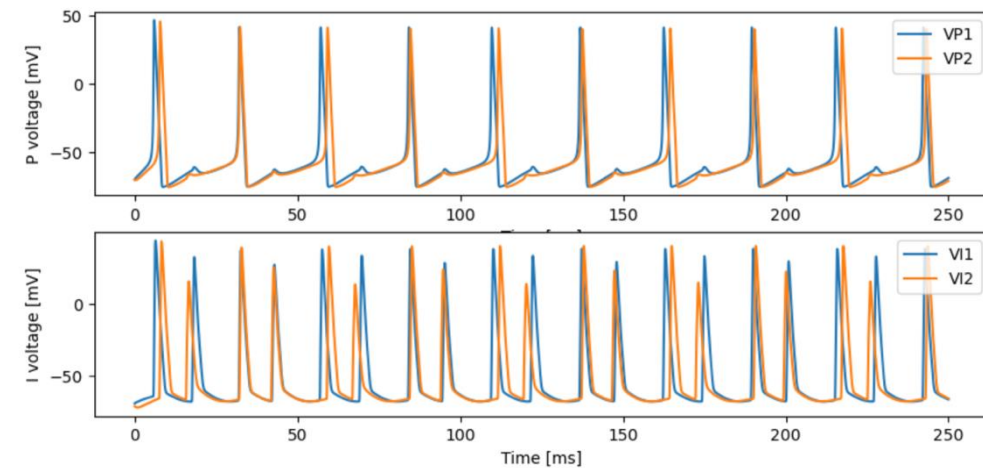
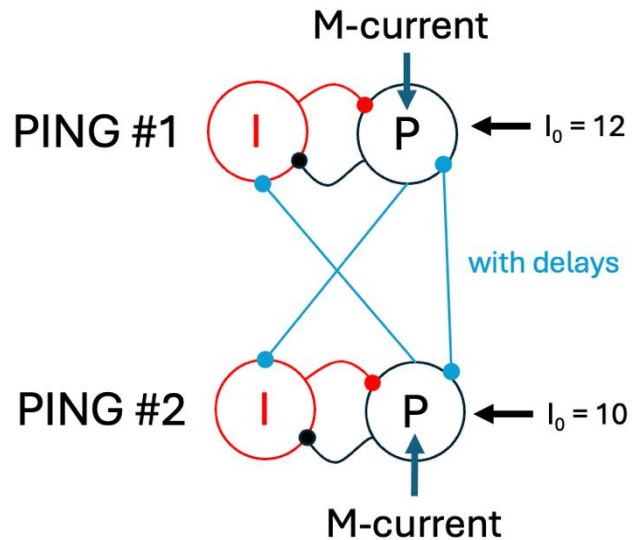
Q: Are the populations synchronized?

Gamma + beta model: Summary

Gamma: local synchrony
(no synaptic delays)



Beta: distant synchrony
(synaptic delays)

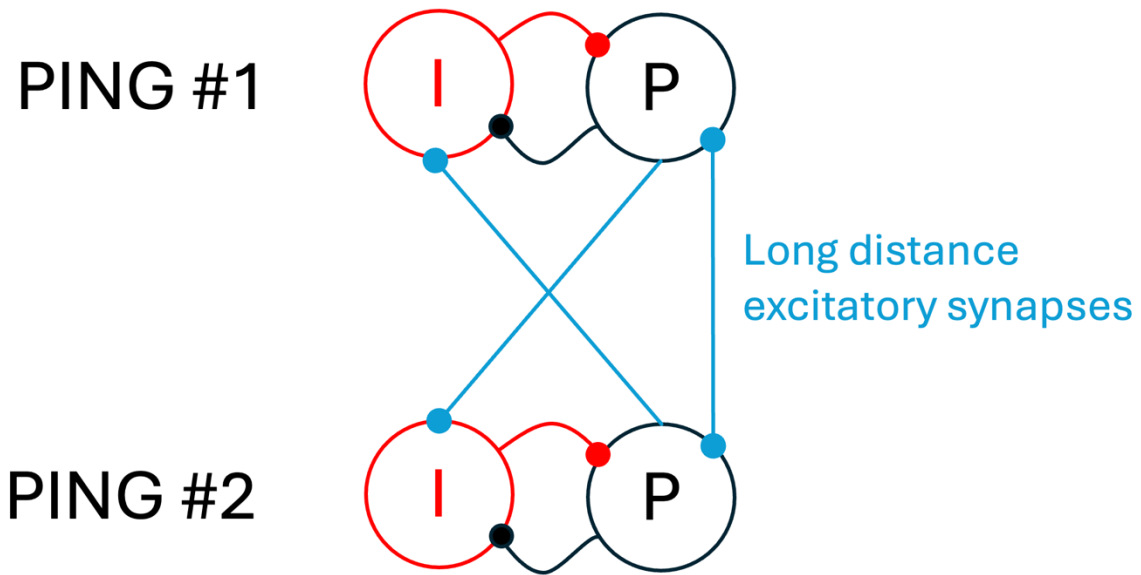


Gamma + beta model: Summary

More generally

High frequencies: local synchrony

Lower frequencies: distant synchrony



Encodes color: **green**

Link (bind) them 

Encodes shape: 

Beta frequency to bind